

PROJECT PROPOSAL • SOFTWARE ENGINEERING

Food Donation Platform

Connecting surplus food from restaurants to NGOs and volunteers — in real time.





The Problem

Every day, restaurants and event venues throw away large quantities of edible, surplus food — while nearby NGOs and shelters struggle to feed people in need.

The gap isn't a lack of food — it's a lack of a fast, reliable way to connect the two sides before the food goes to waste.



1/3

of all food produced globally is wasted



0

easy way to connect restaurants with NGOs today

The Solution

A web & mobile platform that connects three groups in one seamless donation loop:



Restaurants

Post surplus food available for donation, with quantity and pickup window.



NGOs

Browse live listings nearby and claim food donations for their shelters.



Volunteers

Accept pickup & delivery tasks, guided by the live map.

Key Features



Connect Restaurants & NGOs

Restaurants post surplus food; NGOs browse and claim listings instantly, cutting the time food sits unclaimed.



Live Map

Real-time map shows nearby available donations and volunteer locations, so pickups happen fast.



Pickup Tracking

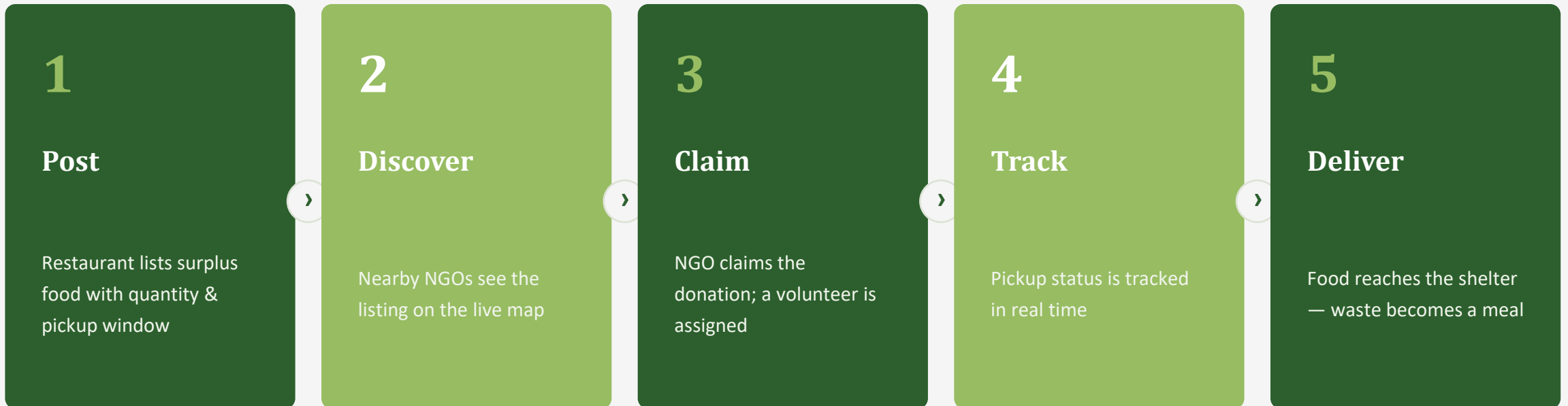
Status updates from claimed to picked up to delivered, so every donation is accounted for.



Notifications & Alerts

Instant alerts to NGOs and volunteers the moment new food is listed nearby.

How It Works



Proposed Technology Stack



Frontend

React.js / Flutter (mobile app) for restaurants, NGOs & volunteers



Backend

Node.js / Django REST API handling listings, claims & tracking



Database

MongoDB / PostgreSQL for users, donations & pickup records



Maps & Location

Google Maps API for live tracking of pickups & routes



Notifications

Firebase Cloud Messaging for real-time alerts



Hosting

AWS / Firebase for deployment & scalability

Impact & Benefits



Reduces Food Waste

Surplus food is redirected before it's discarded



Supports Communities

NGOs & shelters get a steady, reliable food source



Faster Coordination

Live map & tracking cut delays between listing and delivery



Transparency

Every donation is logged and trackable end-to-end



Scalable

Can expand city by city, connecting more restaurants & NGOs



Sustainability

Contributes directly to reducing environmental impact of food waste

Future Scope



AI-based prediction of surplus food quantity & best pickup times



Gamification & rewards for restaurants and volunteers



Integration with government food-safety compliance standards



Multi-language support for wider NGO & volunteer reach



Analytics dashboard showing city-wide food-waste reduction impact

Conclusion

The Food Waste Donation Platform turns a logistics gap into a connected, trackable, and scalable solution — helping restaurants reduce waste while helping NGOs feed more people, faster.

Thank You

Software Engineering — Project Idea Presentation

